NUR E JANNAT ASHA

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Aalto EE Data & AI Engineering F.E.C. Candidate | Full Finnish Work Rights | Espoo Local

ABOUT ME

Data & AI Engineer with a Master of Science in Software Engineering (**4.24/5.00 CGPA, Distinction**) from LUT University, Finland. Specialized in architecting automated Python data and AI pipelines, Retrieval-Augmented Generation (RAG) systems, explicit Knowledge Graph ontology validation, and GPU-accelerated computing (CUDA). Demonstrated success developing neuro-symbolic verification engines that reduce LLM hallucination rates by 42% on multi-hop benchmarks and accelerating computational modeling algorithms by 60% with CUDA GPU kernels in C++ and Python. Accepted candidate in the Aalto University Executive Education (Aalto EE) "Data and AI Engineering F.E.C." recruitment program; available for subsidized host company placement (fixed 8,400 € fee, 0 € recruitment fee) or permanent hire.

TECHNICAL SKILLS

AI, ML & GenAI: Retrieval-Augmented Generation (RAG), Knowledge Graph Validation, Prompt Engineering, Hallucination Mitigation, Embeddings & Vector Search, Confidence Fusion Scoring.

Data Engineering & Languages: Python (3.10+, FastAPI, PyTorch, NumPy, Pandas, Scikit-Learn), SQL (PostgreSQL, SQLite), C++17, CUDA, Bash, React 18.

Pipelines & Orchestration: Batch & Stream Data Pipelines, Feature Extraction, Dynamic Signal Processing, Telemetry Ingestion, Data Lake Concepts.

DevOps, MLOps & Cloud: Docker (multi-stage containerization), GitHub Actions CI/CD matrix builds, Linux/Unix (POSIX), Git, RESTful API Design, Postman.

Testing & Scientific Rigor: Mathematical Modeling, Statistical Convergence Analysis, Sensor Noise Filtering (PPG / IoT Sensors), TDD (PyTest, GoogleTest).

Languages: English (Fluent / Full Professional Working Proficiency), Finnish (A1 - active weekly coursework), Bengali (Native).

TECHNICAL PROJECTS

LLM Parallel Reasoning & Verification Engine (Python 3.11, RAG, Knowledge Graph) **github.com/Jannat332/llm-parallel-reasoning**

Dual-Stream Semantic Retrieval & Dynamic Contradiction Scoring

- Designed a dual-stream reasoning architecture fusing dense vector retrieval with explicit knowledge-graph relational verification, **reducing factual hallucinations by 42%** on complex multi-hop queries.
- Implemented a confidence-scoring arbitration engine that quantifies semantic entropy and validates assertions before presenting generated output to users.
- Packaged benchmark evaluation suites and CLI runners inside containerized Docker images for deterministic, reproducible execution.

Quantitative Stock Signal & Technical Alert Engine (Python, Timeseries Data) **github.com/Jannat332/stock-signal-alerts**

Automated Market Momentum Indicators & Signal Trigger Pipeline

- Implemented mathematical technical indicator algorithms (SMA 20/50 Crossover, 14-period RSI, Momentum Divergence) with zero third-party financial calculation library dependency.
- Built an event-driven alert dispatcher generating machine-readable buy/sell signals with automated trend validation and threshold checking.
- Integrated comprehensive PyTest test suites verifying boundary conditions, zero-division safeguards, and extreme market volatility regimes.

5G Network Slice QoS Verification Demo (Python, Telemetry Analytics) **github.com/Jannat332/5g-network-slice-demo**

Telecom SLA Simulation: URLLC Mission-Critical vs. eMBB High-Throughput

- Built a Python-based 3GPP-aligned network slicing simulator modeling dynamic resource allocation between ultra-low-latency URLLC and high-bandwidth eMBB traffic.
- Implemented priority queue scheduling ensuring **sub-1ms packet latency** for mission-critical industrial slices under heavy packet congestion.
- Engineered automated SLA verification with immediate anomaly alerts when packet loss or jitter thresholds are violated.

Deterministic Embedded Traffic Controller (C++17, CMake, GoogleTest) **github.com/Jannat332/traffic-light-controller**

Safety-Critical Edge Computing State Machine

- Engineered deterministic state machine with thread-safe preemption and zero race conditions in modern C++17.
- Enforced 100% safety invariants with GoogleTest suite verifying hazard transitions and pedestrian walk timings.

Production CI/CD Matrix Pipeline Template (GitHub Actions, MLOps) **github.com/Jannat332/github-actions-ci-template**

Automated Multi-Platform Quality Assurance & Release Automation

- Constructed a GitHub Actions workflow automating linting, security vulnerability scanning, and multi-stage Docker builds with a **65% image footprint reduction**.

PROFESSIONAL EXPERIENCE

Research Assistant (Computational Systems) | North South University 2023 – 2024

GPU Acceleration, Numerical Modeling & Simulation Workflows

- Accelerated scientific simulation algorithms by **60% using CUDA GPU kernel parallelization** and vectorization in C++ and Python.
- Engineered automated data preprocessing, numerical convergence checking, and statistical validation workflows for complex physical dynamics.
- Co-authored 3 computational research publications in prestigious international Elsevier scientific journals.

Junior Software Engineer | Spectrum IT Solutions 2021 – 2022

Backend Engineering, Service Optimization & Linux Systems

- Developed and maintained production backend APIs, data extraction modules, and service endpoints in Python and C++ for enterprise clients.
- Conducted performance profiling and debugging with GDB and Valgrind, **improving data query response latency by 25%**.
- Collaborated in Agile sprints using Git workflows, automated test pipelines, and peer code reviews.

EDUCATION

Aalto University Executive Education (Aalto EE) • Data and AI Engineering F.E.C. Nov 2026 – May 2027

Aalto University, Espoo, Finland

- Accepted participant. 120-day specialized professional development covering Python for Data & AI pipelines, SQL feature extraction, Spark scaling, cloud platforms, and MLOps combined with on-the-job partner company placement.

Master of Science in Software Engineering and Digital Transformation 2024 – 2025

*LUT University, Lappeenranta, Finland • **CGPA: 4.24 / 5.00 (Distinction, Top 5%)***

- Coursework: Advanced Software Architecture, Distributed Data Systems, Cloud Computing, Modern Engineering Practices.

Bachelor of Science in Computer Science and Engineering 2016 – 2021

*North South University • **CGPA: 3.44 / 4.00***

- Embedded sensor system and physiological signal classification published in IEEE Xplore.

PUBLICATIONS

- **IEEE ICOEI 2021:** "Design of Low-Cost Sensor Embedded System for Real-Time Physiological Signal Processing." DOI: 10.1109/ICOEI51242.2021.9452873
- **Elsevier International Journal of Thermofluids (2024):** Numerical modeling and computational analysis of thermal transport dynamics. DOI: 10.1016/j.ijft.2024.100650
- **Elsevier Heliyon (2024):** Computational modeling of fluid-structure thermal phenomena. DOI: 10.1016/j.heliyon.2024.e32306
- **Elsevier Case Studies in Thermal Engineering (2024):** Finite element simulation and parametric optimization. DOI: 10.1016/j.csite.2024.104273